ZYCEVA

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In a double-blind, randomized phase III study (MO22162, CURRENTS) comparing two doses of Erlotinib Tablets (300 mg versus 150 mg) in current smokers (mean of 38 pack years) with locally advanced or metastatic NSCLC in the second-line setting after failure on chemotherapy, the 300 mg dose of Erlotinib Tablets demonstrated no PFS benefit over the recommended dose (7.00 vs 6.86 weeks, respectively). Patients in this study were not selected based on EGFR mutation status.

Pancreatic Cancer (*Erlotinib Tablets administered concurrently with gemcitabine*) The efficacy and safety of Erlotinib Tablets in combination with gemcitabine as a first line treatment was assessed in a randomized, double blind, placebo-controlled trial in 569 patients with locally advanced, unresectable or metastatic pancreatic cancer. Patients were randomized 1:1 to receive Erlotinib Tablets (100 mg or 150 mg) or placebo once daily on a continuous schedule plus gemcitabine IV (1000 mg/m2, Cycle 1 - Days 1, 8, 15, 22, 29, 36 and 43 of an 8 week cycle; Cycle 2 and subsequent cycles - Days 1, 8 and 15 of a 4 week cycle [approved dose and schedule for pancreatic cancer, see the gemcitabine SPC]). Erlotinib Tablets or placebo was taken orally once daily until disease progression or unacceptable toxicity. Study end points included overall survival, response rate and PFS.

Duration of response was also examined. The primary endpoint was survival. A total of 285 patients were randomized to receive gemcitabine plus Erlotinib Tablets (261 patients in the 100 mg cohort and 24 patients in the 150 mg cohort) and 284 patients were randomized to receive gemcitabine plus placebo (260 patients in the 100 mg cohort and 24 patients in the 150 mg cohort). Too few observations were made for the 150 mg cohort to draw conclusions.

Baseline demographic and disease characteristics of the patients were similar between the 2 treatment groups, 100 mg Erlotinib Tablets plus gemcitabine or placebo plus gemcitabine, except for a slightly larger proportion of females in the Erlotinib Tablets arm (51%) compared with the placebo arm (44%). The median time from initial diagnosis to randomization was approximately 1.0 month. Approximately half of the patients had a baseline ECOG performance status (PS) of 1, and 17 % had a baseline ECOG PS of 2. Most patients presented with metastatic disease at study entry as the initial manifestation of pancreatic cancer (77% in the Erlotinib Tablets arm, 76% in the placebo arm).

Survival was evaluated in the intent-to-treat population based on follow-up survival data including 551 deaths. Results are presented for the 100 mg dose cohort (504 deaths). The adjusted hazard ratio for death in the Erlotinib Tablets group relative to the placebo group was 0.82 (95 % CI, 0.69 to 0.98) (p = 0.028). The percent of patients alive at 12 months was 23.8 % in the Erlotinib Tablets group compared to 19.4% in the placebo group. The median overall survival was 6.4 months in the Erlotinib Tablets group compared with 6 months in the placebo group. Table 4 summarizes the results of the study.

Table 4: Study PA3 Efficacy Results

	Erlotinib Tablets 100mg plus gemcitabine (N=261)	Placebo plus gemcitabine (N=260)	p-value
Median survival	6.4 months	6 months	
Hazard ratio, mortality (erlotinib:placebo) (95%CI)	0.82 (0.69 to 0.98)		p = 0.028
% Patients alive at 12 months	23.8	19.4	

The median PFS was 3.81 months (16.5 weeks) in the Erlotinib Tablets group (95 % CI, 3.58 to 4.93 months) compared with 3.55 months (15.2 weeks) in the placebo group (95 % CI, 3.29 to 3.75 months) (p = 0.006).

The median duration of response was 23.9 weeks, ranging from 3.71 to 56+ weeks. The objective response rate (complete response and partial response) was 8.6 % in the Erlotinib Tablets group and 7.9% in the placebo group. The proportion of patients who experienced complete response, partial response or stable disease was 59 % and 49.4 %, respectively, for the Erlotinib Tablets and placebo groups (p = 0.036).

3.2 Pharmacokinetic Properties

Exposure:

Following a 150 mg oral dose of Erlotinib Tablets, at steady state, the median time to reach maximum plasma concentrations is approximately 4.0 hours with median maximum plasma concentrations achieved of 1,995 ng/ml. Prior to the next dose at 24 hours, the median minimum plasma concentrations are 1,238 ng/ml. Median AUC achieved during the dosing interval at steady state are 41.300 mcg*hr/ml.

3.2.1 Absorption

Oral Erlotinib is well absorbed and has an extended absorption phase, with mean peak plasma levels occurring at 4 hours after oral dosing. A study in normal healthy volunteers provided an estimate of bioavailability of 59%. The exposure after an oral dose may be increased by food. Following absorption, Erlotinib is highly bound in blood, with approximately 95% bound to blood components, primarily to plasma proteins (i.e. albumin and alpha-1 acid glycoprotein [AAG]), with a free fraction of approximately 5%.

3.2.2 Distribution

Erlotinib has a mean apparent volume of distribution of 232 L and distributes into tumor tissue of humans. In a study of 4 patients (3 NSCLC, and 1 with laryngeal cancer) receiving 150 mg daily oral doses of Erlotinib Tablets, tumor samples from surgical excisions on Day 9 of treatment revealed tumor concentrations of erlotinib that averaged 1,185 ng/g of tissue. This corresponded to an overall average of 63% of the steady state observed peak plasma concentrations. The primary active metabolites were present in tumor at concentrations averaging 160 ng/g tissue, which corresponded to an overall average of 113% of the observed steady state peak plasma concentrations. Tissue distribution studies using whole body autoradiography following oral administration with [14C] labeled erlotinib in athymic nude mice with HN5 tumor xenografts have shown rapid and extensive tissue distribution with maximum concentrations of radiolabeled drug (approximately 73% of that in plasma) observed at 1 hour.

3.2.3 Metabolism

Erlotinib is metabolised in humans by hepatic cytochrome P450 enzymes, primarily by CYP3A4 and to a lesser extent by CYP1A2. Extrahepatic metabolism by CYP3A4 in intestine, CYP1A1 in lung, and CYP1B1 in tumour tissue potentially contribute to the metabolic clearance of erlotinib. *In vitro* studies indicate approximately 80-95% of erlotinib metabolism by the CYP3A4 enzyme. There are three main metabolic pathways identified: 1) O-demethylation of either side chain or both, followed by oxidation to the carboxylic acids; 2) oxidation of the acetylene moiety followed by hydrolysis to the aryl carboxylic acid; and 3) aromatic hydroxylation of the phenyl-acetylene moiety. The primary metabolites of erlotinib produced by O-demethylation of either side chain have comparable potency to erlotinib in nonclinical *in vitro* assays and *in vivo* tumor models.

They are present in plasma at levels that are <10% of erlotinib and display similar pharmacokinetics as erlotinib.

3.2.4 Elimination

The metabolites and trace amounts of erlotinib are excreted predominantly via the feces (>90%), with renal elimination accounting for only a small amount of an oral dose.

Clearance:

A population pharmacokinetic analysis in 591 patients receiving single agent Erlotinib Tablets show a mean apparent clearance of 4.47 L/hour with a median half-life of 36.2 hours. Therefore, the time to reach steady state plasma concentration would be expected to occur in approximately 7-8 days. No significant relationships between predicted apparent clearance and patient age, body weight, gender, and ethnicity were observed. Patient factors, which correlate with erlotinib pharmacokinetics, are serum total bilirubin, AAG concentrations and current smoking. Increased serum concentrations of total bilirubin and AAG concentrations were associated with a slower rate of erlotinib clearance. Smokers had a higher rate of erlotinib clearance (See 2.8, Interactions with other medicinal products and other forms of interaction).

A second population pharmacokinetic analysis was conducted that incorporated erlotinib data from 204 pancreatic cancer patients who received erlotinib plus gemcitabine. This analysis demonstrated that covariates affecting erlotinib clearance in patients from the pancreatic study were very similar to those seen in the prior single-agent pharmacokinetic analysis. No new covariate effects were identified. Co-administration of gemcitabine had no effect on erlotinib plasma clearance.

3.2.5 Pharmacokinetics in Special Populations

There have been no specific studies in pediatric or elderly patients.

Hepatic impairment: Erlotinib is mainly cleared by the liver. Erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function including patients with primary liver cancer or hepatic metastases.

Renal impairment: Erlotinib and its metabolites are not significantly excreted by the kidneys, as less than 9% of a single dose is excreted in the urine. No clinical studies have been conducted in patients with compromised renal function.

Smokers: A pharmacokinetic study in nonsmoking and currently cigarette smoking healthy subjects has shown that cigarette smoking leads to increased clearance of, and decreased exposure to, erlotinib. The AUC0-infinity in smokers was about 1/3 of that in never/former smokers (n=16 in each of smoker and never/former smoker arms). This reduced exposure in current smokers is presumably due to induction of CYP1A1 in lung and CYP1A2 in the liver.

In the pivotal Phase III NSCLC trial, current smokers achieved erlotinib steady state trough plasma concentration of 0.65 µg/mL (n=16) which was approximately 2-fold less than the former smokers or patients who had never smoked (1.28 µg/mL, n=108). This effect was accompanied by a 24% increase in apparent erlotinib plasma clearance.

In a phase I dose escalation study in NSCLC patients who were current smokers, pharmacokinetic analyses at steady-state indicated a dose proportional increase in erlotinib exposure when the Erlotinib Tablets dose was increased from 150 mg to the maximum tolerated dose of 300 mg. Steady-state trough plasma concentrations at a 300mg dose in current smokers in this study was 1.22 µg/mL (n=17) (See sections 2.2.2 Special Dosage Instructions, 2.8, Interactions with other Medicinal Products and other Forms of Interaction).

4. PHARMACEUTICAL PARTICULARS

4.1 Storage

Erlotinib Tablets should not be used after the expiry date (EXP) shown on the pack.
Do not store above 30°C.

4.2 Special Instructions for Use, Handling and Disposal

Disposal of unused/expired medicines

The release of pharmaceuticals in the environment should be minimized. Medicines should not be disposed of via wastewater and disposal through household waste should be avoided. Any unused medicinal product or waste material should be disposed of in accordance with local requirements

4.3 Packs

Blister pack of 3 x 10 Tablets

Medicine: keep out of reach of children

Product Registration Holder:
Healol Pharmaceuticals Sdn. Bhd.,
No. 74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju, 53300 Kuala Lumpur, Malaysia.

Dedicated To Life

Manufactured by:
Zydus Lifesciences Limited,
Plot No. - 1- A / 1 & 2,
Pharmez, (Special Economic Zone), Matoda,
Sarkhej Bavla N. H. No. 8A, Tal. - Sanand,
Dist. - Ahmedabad - 382 213, Gujarat State, INDIA.

Date of Revision:
13/07/2024

1. DESCRIPTION

1.1 Therapeutic / Pharmacologic Class of Drug

Antineoplastic agent.
ATC code: L01EB02 – ERLOTINIB

1.2 Type of Dosage Form

Film-coated tablets

ZYCEVA 100
Erlotinib Film Coated Tablets 100mg
White to off-white, round, film coated tablets, debossed with '914' on one side and plain on other side.

ZYCEVA 150
Erlotinib Film Coated Tablets 150mg
White to off-white, round, film coated tablets, debossed with '915' on one side and plain on other side.

1.3 Route of Administration

Oral.

1.4 Qualitative and Quantitative Composition

Active ingredient: Erlotinib hydrochloride.
One film-coated tablet of each strength contains erlotinib hydrochloride, corresponding to 100 mg and 150 mg of erlotinib.

List of excipient
Tablet core
Lactose Monohydrate, Microcrystalline Cellulose 101, Sodium lauryl Sulfate, Crospovidone (Type-A), Hypromellose 6 CPS, 2910, Microcrystalline Cellulose 102, Crospovidone (Type-B), Colloidal Silicon Dioxide, Magnesium Stearate

Tablet coat
Opadry White 03F580010 (HPMC 2910/Hypromellose, Titanium Dioxide, Macrogol/PEG, Talc)

2. CLINICAL PARTICULARS

2.1 Therapeutic Indication(s)

Non-small cell lung cancer

Erlotinib Tablets is indicated for the first-line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with EGFR activating mutations.

Erlotinib Tablets is indicated for maintenance treatment in patients with locally advanced or metastatic NSCLC with EGFR activating mutations who have not progressed after first-line chemotherapy.

Erlotinib Tablets is also indicated for the treatment of patients with locally advanced or metastatic NSCLC after failure of at least one prior chemotherapy regimen.

Pancreatic Cancer

Erlotinib Tablets in combination with gemcitabine is indicated for the treatment of patients with locally advanced, unresectable or metastatic pancreatic cancer

2.2 Dosage and Administration

2.2.1 Standard Dosage

Non-small cell lung cancer

EGFR mutation testing should be performed prior to initiation of Erlotinib Tablets as first-line or maintenance therapy in patients with locally advanced or metastatic NSCLC.

The recommended daily dose of Erlotinib Tablets is 150 mg taken at least one hour before or two hours after the ingestion of food.

Pancreatic Cancer

The recommended daily dose of Erlotinib Tablets is 100 mg taken at least one hour before or two hours after the ingestion of food, in combination with gemcitabine (see the package insert of gemcitabine for the pancreatic cancer indication)

2.2.2 Special Dosage Instructions

Drug Interactions: Concomitant use of CYP 3A4 substrates and modulators may require dose adjustment (see section 2.8, Interactions with other Medicinal Products and other Forms of Interaction).

When dose adjustment is necessary, it is recommended to reduce in 50 mg steps (see sections 2.4, Warnings and Precautions and 2.8, Interactions with other Medicinal Products and other Forms of Interaction).

Hepatic impairment: Erlotinib is eliminated by hepatic metabolism and biliary excretion. Although erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function, caution should be used when administering Erlotinib Tablets to patients with hepatic impairment. Dose reduction or interruption of Erlotinib Tablets should be considered if severe adverse reactions occur. Safety and efficacy have not been studied in patients with severe hepatic impairment (see sections 2.4.1, Warnings & Precautions [*Hepatitis, hepatic failure*] and 3.2.5, Pharmacokinetics in Special Populations)

Renal impairment: The safety and efficacy of Erlotinib Tablets has not been studied in patients with renal impairment. (see section 3.2.5, Pharmacokinetics in Special Populations).

Pediatric use: The safety and efficacy of Erlotinib Tablets in the approved indications has not been established in patients under the age of 18 years.

Smokers: Cigarette smoking has been shown to reduce erlotinib exposure by 50-60%.

The maximum tolerated dose of Erlotinib Tablets in NSCLC patients who currently smoke cigarettes was 300 mg. The 300mg dose did not show improved efficacy in second line treatment after failure of chemotherapy compared to the recommended 150mg dose in patients who continue to smoke cigarettes (see section 2.8, Interactions with other Medicinal Products and other Forms of Interaction and section 3.2.5 Pharmacokinetics in Special Populations).

2.3 Contraindications

Erlotinib Tablets is contraindicated in patients with severe hypersensitivity to erlotinib or to any component of Erlotinib Tablets.

2.4 Warnings and Precautions

2.4.1 General

Interstitial Lung Disease: Cases of interstitial lung disease (ILD)-like events, including fatalities, have been reported uncommonly in patients receiving Erlotinib Tablets for treatment of NSCLC (non-small cell lung carcinoma), pancreatic cancer or other advanced solid tumors. In pivotal study BR 21, in NSCLC, the incidence of serious ILD-like events was 0.8% in each of the placebo and Erlotinib Tablets arms. In a meta-analysis of NSCLC randomized controlled clinical trials, the incidence of ILD-like events was 0.9% on Erlotinib Tablets compared to 0.4% in patients in the control arms in the pancreatic cancer study in combination with gemcitabine, the incidence of ILD-like events was 2.5% in the Erlotinib Tablets plus gemcitabine group versus 0.4% in the placebo plus gemcitabine treated group.

Some examples of reported diagnoses in patients suspected of having ILD-like events include pneumonitis, radiation pneumonitis, hypersensitivity pneumonitis, interstitial pneumonia, interstitial lung disease, obliterative bronchiolitis, pulmonary fibrosis, Acute Respiratory Distress Syndrome, lung infiltration and alveolitis. These ILD-like events started from a few days to several months after initiating Erlotinib therapy. Most of the cases were associated with confounding or contributing factors such as concomitant or prior chemotherapy, prior radiotherapy, pre-existing parenchymal lung disease, metastatic lung disease, or pulmonary infections.

In patients who develop acute onset of new and/or progressive unexplained pulmonary symptoms, such as dyspnea, cough and fever, Erlotinib therapy should be interrupted pending diagnostic evaluation. If ILD is diagnosed, Erlotinib should be discontinued and appropriate treatment initiated as necessary (see section 2.6, Undesirable Effects).

Diarrhea, Dehydration, Electrolyte Imbalance and Renal failure: Diarrhea has occurred in patients on Erlotinib Tablets and moderate or severe diarrhea should be treated with loperamide. In some cases, dose reduction may be necessary. In the event of severe or persistent diarrhea, nausea, anorexia, or vomiting associated with dehydration, Erlotinib therapy should be interrupted and appropriate measures should be taken to treat the dehydration (see section 2.6, Undesirable Effects). There have been rare reports of hypokalemia and renal failure (including fatalities). Some reports of renal failure were secondary to severe dehydration due to diarrhea, vomiting and/or anorexia while others were confounded by concomitant chemotherapy. In more severe or persistent cases of diarrhea, or cases leading to dehydration, particularly in groups of patients with aggravating risk factors (concomitant medications, symptoms or diseases or other predisposing conditions including advanced age), Erlotinib therapy should be interrupted and appropriate measures should be taken to intensively rehydrate the patients intravenously. In addition, renal function and serum electrolytes including potassium should be monitored in patients at risk of dehydration.

Hepatitis, hepatic failure: Rare cases of hepatic failure (including fatalities) have been reported during use of Erlotinib Tablets. Confounding factors have included pre-existing liver disease or concomitant hepatotoxic medications. Therefore, in such patients, periodic liver function testing should be considered. Erlotinib Tablets dosing should be interrupted if changes in liver function are severe (see section 2.6, Undesirable Effects)

Gastrointestinal Perforation: Patients receiving Erlotinib Tablets are at increased risk of developing gastrointestinal perforation, which was observed uncommonly (including some cases with a fatal outcome). Patients receiving concomitant anti-angiogenic agents, corticosteroids, NSAIDs, and/or taxane based chemotherapy, or who have prior history of peptic ulceration or diverticular disease are at increased risk. Erlotinib Tablets should be permanently discontinued in patients who develop gastrointestinal perforation (see section 2.6, Undesirable Effects).

***Bullous and exfoliative skin disorders*:** Bullous, blistering and exfoliative skin conditions have been reported, including very rare cases suggestive of Stevens-Johnson syndrome/Toxic epidermal necrolysis, which in some cases were fatal (see section 2.6, Undesirable Effects). Erlotinib Tablets treatment should be interrupted or discontinued if the patient develops severe bullous, blistering or exfoliating conditions.

Ocular Disorders: Very rare cases of corneal perforation or ulceration have been reported during use of Erlotinib Tablets. Other ocular disorders including abnormal eyelash growth, keratoconjunctivitis sicca or keratitis have been observed with Erlotinib Tablets treatment which are also risk factors for corneal perforation/ulceration. Erlotinib therapy should be interrupted or discontinued if patients present with acute/worsening ocular disorders such as eye pain (see section 2.6, Undesirable Effects).

Drug Interactions: Erlotinib has a potential for clinically significant drug-drug interactions. (See section 2.8, Interactions with other Medicinal Products and other Forms of Interaction)

2.4.2 Ability to Drive and Use Machines

Erlotinib has no or negligible influence on the ability to drive and use machines.

2.5 Use in Special Populations

2.5.1 Potential Females and Males of Reproductive Potential

Contraception: Females:

Adequate contraceptive methods should be used during therapy, and for at least 2 weeks after completing therapy.

2.5.2 Pregnancy

There are no adequate or well controlled studies in pregnant women using Erlotinib Tablets.
Studies in animals have shown some reproductive toxicity (see sections 3..3.3, Impairment of Fertility and 3.3.4, Reproductive toxicity). The potential risk for humans is unknown. Women of childbearing potential must be advised to avoid pregnancy while on Erlotinib Tablets. Treatment should only be continued in pregnant women if the potential benefit to the mother outweighs the risk to the fetus.

2.5.3 Lactation

Nursing mothers: It is not known whether erlotinib is excreted in human milk. No studies have been conducted to assess the impact of Erlotinib Tablets on milk production or its presence in breast milk. As the potential for harm to the nursing infant is unknown, mothers should be advised against breastfeeding while receiving Erlotinib Tablets and for at least 2 weeks after the final dose.

2.5.4 Pediatric Use

The safety and efficacy of Erlotinib Tablets in the approved indications has not been established in patients under the age of 18 years (see sections 2.2.2, Special Dosage Instructions and 3.2.5 Pharmacokinetics in Special Population).

2.5.5 Geriatric Use

See section 3.2.5, Pharmacokinetics in Special Population.

2.5.6 Renal Impairment

See sections 2.2.2, Special Dosage Instructions and 3.2.5, Pharmacokinetics in Special Population.

2.5.7 Hepatic Impairment

Erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function including patients with primary liver cancer or hepatic metastases (see section 2.4.1 Warnings and Precautions). Safety and efficacy have not been studied in patients with severe hepatic impairment (see section 2.2.2, Special Dosage Instructions).

2.6 Undesirable Effects

2.6.1 Clinical Trials

Safety evaluation of Erlotinib Tablets is based on the data from more than 1500 patients treated with at least one 150 mg dose of Erlotinib Tablets monotherapy, and more than 300 patients who received Erlotinib Tablets 100 mg or 150 mg in combination with gemcitabine.

The incidence of adverse drug reactions (ADRs) reported with Erlotinib Tablets alone or in combination with chemotherapy are summarized in the tables below and are based on data from clinical trials. The listed ADRs were those reported in at least 10% (in the Erlotinib Tablets group) of patients and occurred more frequently (≥3%) in patients treated with Erlotinib Tablets than in the comparator arm.

600 mm

180 mm

Adverse drug reactions from clinical trials (Table 1) are listed by MedDRA system organ class. The corresponding frequency category for each adverse drug reaction is based on the following convention: very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1000), very rare (<1/10,000).

Non-small cell lung cancer - Erlotinib Tablets administered as monotherapy

First-Line Treatment of Patients with EGFR Mutations
In an open-label, randomized phase III study, ML 20650 conducted in 154 patients, the safety of Erlotinib Tablets for first-line treatment of NSCLC patients with EGFR activating mutations was assessed in 75 patients; no new safety signals were observed in these patients. The most frequent ADRs seen in patients treated with Erlotinib Tablets in study ML 20650 were rash and diarrhoea (any Grade 80% and 57%, respectively), most were Grade 1/2 in severity and manageable without intervention. Grade 3 rash and diarrhea occurred in 9% and 4% of patients, respectively. No Grade 4 rash or diarrhoea was observed. Both rash and diarrhoea resulted in discontinuation of Erlotinib Tablets in 1% of patients. Dose modifications (interruptions or reductions) for rash and diarrhoea were needed in 11% and 7% of patients, respectively.

Maintenance treatment
In two other double-blind, randomized, placebo-controlled Phase III studies (BO18192 and BO25460) conducted in a total of 1532 patients with advanced, recurrent or metastatic NSCLC following first-line standard platinum-based chemotherapy, no new safety signals were identified. The most frequent ADRs seen in patients treated with Erlotinib Tablets in studies BO18192 and BO25460 were rash (BO18192: all grades 49.2%, grade 3: 6.0%; BO25460: all grades 39.4%, grade 3: 5.0%) and diarrhea (BO18192: all grades 20.3%, grade 3: 1.8%; BO25460: all grades 24.2%, grade 3: 2.5%).

No Grade 4 rash or diarrhoea was observed in either study. Rash and diarrhoea resulted in discontinuation of Erlotinib Tablets in 1% and < 1% of patients, respectively, in study BO18192, while no patient discontinued for rash or diarrhea in BO25460. Dose modifications (interruptions or reductions) for rash and diarrhoea were needed in 8.3% and 3% of patients, respectively, in study BO18192 and 5.6% and 2.8% of patients, respectively, in study BO25460.

Second and Further Line Treatment
The ADRs in Table 1 are based on data from a randomized double-blind study (BR.21) conducted in 731 patients with locally advanced or metastatic NSCLC after failure of at least one prior chemotherapy regimen. Patients were randomized 2:1 to receive Erlotinib Tablets 150 mg or placebo. Study drug was taken orally once daily until disease progression or unacceptable toxicity.

The most frequent ADRs were rash and diarrhea (any Grade 75% and 54% respectively), most were Grade 1/2 in severity and manageable without intervention. Grade 3/4 rash and diarrhea occurred in 9% and 6%, respectively, in patients treated with Erlotinib Tablets and each resulted in study discontinuation in 1% of patients. Dose reduction for rash and diarrhea was needed in 6% and 1% of patients, respectively. In study BR.21, the median time to onset of rash was 8 days, and the median time to onset of diarrhea was 12 days.

Pancreatic cancer- Erlotinib Tablets administered concurrently with gemcitabine
The ADRs listed in the Table 1 below are based on erlotinib-arm data from a controlled clinical trial (PA.3), 259 patients with pancreatic cancer received Erlotinib Tablets 100mg plus gemcitabine compared to 256 patients in the placebo plus gemcitabine-arm.

The most frequent ADRs in study PA.3 in pancreatic cancer patients receiving Erlotinib Tablets 100 mg plus gemcitabine were fatigue, rash and diarrhea. In the Erlotinib Tablets plus gemcitabine arm, Grade 3/4 rash and diarrhea were reported in 5% of patients. The median time to onset of rash and diarrhea was 10 days and 15 days, respectively. Rash and diarrhea each resulted in dose reductions in 2% of patients, and resulted in study discontinuation in up to 1% of patients receiving Erlotinib Tablets plus gemcitabine.

The Erlotinib Tablets 150 mg plus gemcitabine cohort (23 patients) was associated with a higher rate of certain class-specific adverse reactions including rash and required more frequent dose reduction or interruption.

Table 1 ADRs occurring in ≥ 10% of patients in BR.21 (treated with Erlotinib Tablets) and PA.3 (treated with Erlotinib Tablets plus gemcitabine) studies and ADRs occurring more frequently (≥ 3%) than placebo in BR.21 (treated with Erlotinib Tablets) and PA.3 (treated with Erlotinib Tablets plus gemcitabine) studies

	Erlotinib Tablets (BR.21) N = 485			Erlotinib Tablets (PA.3) N = 259			Frequency category of highest incidence
NCI-CTC Grade	Any Grade	3	4	Any Grade	3	4	
MedDRA Preferred Term	%	%	%	%	%	%	
Infections and infestations							
Infection*	24	4	0	31	3	<1	very common
Metabolism and nutrition disorders							
Anorexia	52	8	1	-	-	-	very common
Weight decreased	-	-	-	39	2	0	very common
Eye disorders							
Keratoconjunctivitis sicca	12	0	0	-	-	-	very common
Conjunctivitis	12	<1	0	-	-	-	very common
Psychiatric disorders							
Depression	-	-	-	19	2	0	very common
Nervous system disorders							
Headache	-	-	-	13	1	<1	very common
Neuropathy	-	-	-	15	<1	0	very common
Respiratory, thoracic and mediastinal disorders							
Dyspnoea	41	17	11	-	-	-	very common
Cough	33	4	0	16	0	0	very common
Gastrointestinal disorders							
Diarrhoea**	54	6	<1	48	5	<1	very common
Nausea	33	3	0	-	-	-	very common
Vomiting	23	2	<1	-	-	-	very common
Stomatitis	17	<1	0	22	<1	0	very common
Abdominal pain	11	2	<1	-	-	-	very common
Dyspepsia	-	-	-	17	<1	0	very common
Flatulence	-	-	-	13	0	0	very common
Skin and subcutaneous tissue disorders							
*Rash****	75	8	<1	69	5	0	very common
Pruritus	13	<1	0	-	-	-	very common
Dry skin	12	0	0	-	-	-	very common
Alopecia	-	-	-	14	0	0	very common
General disorders and administration site conditions							
Fatigue	52	14	4	73	14	2	very common
Pyrexia	-	-	-	36	3	0	very common
Rigors	-	-	-	12	0	0	very common

* Severe infections, with or without neutropenia, have included pneumonia, sepsis, and cellulitis.
** Can lead to dehydration, hypokalemia and renal failure.
*** Rash included dermatitis acneiform.
\- corresponds to percentage below threshold.

Further information on ADRs of special interest
The following ADRs have been observed in patients who received Erlotinib Tablets 150 mg as monotherapy or 100 mg or 150 mg in combination with gemcitabine.
Very common ADR's are presented in Table 1, ADRs in other frequency categories are summarized below.

Gastrointestinal disorders:
Gastrointestinal perforations have been reported uncommonly (in less than 1% of patients) with Erlotinib Tablets treatment, in some cases with a fatal outcome (see section 2.4 Warnings and Precautions).
Cases of gastrointestinal bleeding have been reported commonly (including some fatalities), some associated with concomitant warfarin administration (see also section 2.8, Interactions with other Medicinal Products and other Forms of Interaction), and some with concomitant NSAIDs.

Hepatobiliary disorders:
Liver function test abnormalities (including raised ALT, AST, bilirubin) have been observed commonly in clinical trials of Erlotinib Tablets. In study PA3, these occurred very commonly. These were mainly mild or moderate in severity, transient in nature or associated with liver metastases. Rare cases of hepatic failure (including fatalities) have been reported during use of Erlotinib Tablets. Confounding factors have included pre-existing liver disease or concomitant hepatotoxic medications (see section 2.4, Warnings and Precautions).

Eye disorders:
Corneal ulcerations or perforations have been reported very rarely in patients receiving Erlotinib Tablets treatment (see section 2.4 Warnings and Precautions). Keratitis and conjunctivitis has been reported commonly with Erlotinib Tablets. Abnormal eyelash growth including; in-growing eyelash, excessive growth and thickening of the eyelashes have been reported (see section 2.4 Warnings and Precautions).

Respiratory, thoracic and mediastinal disorders:
There have been uncommon reports of serious interstitial lung disease (ILD)-like events, (including fatalities) in patients receiving Erlotinib Tablets for treatment of NSCLC and other advanced solid tumors (see section 2.4, Warnings and Precautions).

Cases of epistaxis have also been reported commonly in both the NSCLC and the pancreatic cancer trial.

Skin and subcutaneous tissue disorders:
Rash has been reported very commonly in patients receiving Erlotinib Tablets and in general, manifests as a mild or moderate erythematous and papulopustular rash, which may occur or worsen in sun exposed areas. For patients who are exposed to sun, protective clothing, and/or use of sun screen (e.g. mineral-containing) may be advisable. Acne, dermatitis acneiform and folliculitis have been observed commonly, most of these events were mild or moderate and non-serious. Skin fissures, mostly non-serious, were reported commonly and in the majority of cases were associated with rash and dry skin. Other mild skin reactions such as hyperpigmentation have been observed uncommonly (in less than 1% of patients).

Bullous, blistering and exfoliative skin conditions have been reported, including very rare cases suggestive of Stevens-Johnson syndrome/Toxic epidermal necrolysis, which in some cases were fatal (see section 2.4 Warnings and Precautions).

Hair and nail changes, mostly non-serious, were reported in clinical trials, e.g. paronychia was reported commonly and hirsutism, eyelash/eyebrow changes and brittle and loose nails were reported uncommonly.

2.6.2 Post Marketing Experience
The following adverse drug reactions have been identified from postmarketing experience with Erlotinib Tablets based on spontaneous case reports and literature cases.

Table 2 Adverse drug reactions from postmarketing experience

Adverse reactions	Frequency Category	CDS reference
Eye disorders		
Uveitis	Unknown	101
Skin and subcutaneous tissue disorders		
Hair and nail changes, mostly nonserious, e.g. hirsutism, eyelash/eyebrow changes, paronychia and brittle and loose nails	Uncommon	77, 78

2.7 Overdose
Single oral doses of Erlotinib Tablets up to 1000 mg in healthy subjects and up to 1600 mg given as a single dose once weekly in cancer patients, have been tolerated. Repeated twice daily doses of 200 mg in healthy subjects were poorly tolerated after only a few days of dosing. Based on the data from these studies, severe adverse events such as diarrhea, rash, and possibly liver transaminase elevation may occur above the recommended dose of 150 mg. In case of suspected overdose Erlotinib Tablets should be withheld and symptomatic treatment administered.

2.8 Interactions with other Medicinal Products and other Forms of Interaction
Erlotinib is metabolized in the liver by the hepatic cytochromes in humans, primarily CYP3A4 and to a lesser extent by CYP1A2, and the pulmonary isoform CYP1A1.

Potential interactions may occur with drugs which are metabolized by, or are inhibitors or induces of, these enzymes.

Potent inhibitors of CYP3A4 activity decrease erlotinib metabolism and increase erlotinib plasma concentrations. Inhibition of CYP3A4 metabolism by ketoconazole (200 mg po BID for 5 days) resulted in increased exposure to erlotinib (86% in median erlotinib exposure [AUC]) and a 69% increase in Cmax when compared to erlotinib alone.

When Erlotinib Tablets was co-administered with ciprofloxacin, an inhibitor of both CYP3A4 and CYP1A2, the erlotinib exposure [AUC] and maximum concentration (Cmax) increased by 39% and 17%, respectively. Therefore caution should be used when administering Erlotinib Tablets with potent CYP3A4 or combined CYP3A4/CYP1A2 inhibitors.

In these situations, the dose of Erlotinib Tablets should be reduced if toxicity is observed. Potent inducers of CYP34A activity increase erlotinib metabolism and significantly decrease erlotinib plasma concentrations.

Induction of CYP3A4 metabolism by rifampicin (600 mg po QD for 7 days) resulted in a 69% decrease in the median erlotinib AUC, following a 150 mg dose of Erlotinib Tablets as compared to Erlotinib Tablets alone.

Pre-treatment and co-administration of rifampicin with a single 450 mg dose of Erlotinib Tablets resulted in a mean erlotinib exposure [AUC] of 57.5% of that after a single 150 mg Erlotinib Tablets dose in the absence of rifampicin treatment. Alternative treatments lacking potent CYP3A4 inducing activity should be considered when possible. For patients who require concomitant treatment with Erlotinib Tablets and a potent CYP3A4 inducer such as rifampicin an increase in dose to 300 mg should be considered while their safety (see section 2.4.1 Warnings and Precautions, General) is closely monitored, and if well tolerated for more than 2 weeks, further increase to 450 mg could be considered with close safety monitoring. Higher doses have not been studied in this setting.

Pre-treatment or co-administration of Erlotinib Tablets did not alter the clearance of the prototypical CYP3A4 substrates midazolam and erythromycin. Significant interactions with the clearance of other CYP3A4 substrates are therefore unlikely. Oral availability of midazolam did appear to decrease by up to 24%, which was however not attributed to effects on CYP3A4 activity.

The solubility of erlotinib is pH dependent. Erlotinib solubility decreases as pH increases. Drugs that alter the pH of the upper GI tract may alter the solubility of erlotinib and hence its bioavailability. Co-administration of Erlotinib Tablets with omeprazole, a proton pump inhibitor, decreased the erlotinib exposure [AUC] and Cmax by 46% and 61%, respectively. There was no change to Tmax or half-life. Concomitant administration of Erlotinib Tablets with 300 mg ranitidine, an H2-receptor antagonist, decreased erlotinib exposure [AUC] and Cmax by 33% and 54%, respectively. Therefore, co-administration of drugs reducing gastric acid production with Erlotinib Tablets should be avoided where possible. Increasing the dose of Erlotinib Tablets when co-administered with such agents is not likely to compensate for this loss of exposure. However, when Erlotinib Tablets was dosed in a staggered manner 2 hours before or 10 hours after ranitidine 150 mg b.i.d., erlotinib exposure [AUC] and Cmax decreased only by 15% and 17%, respectively. If patients need to be treated with such drugs, then an H2-receptor antagonist such as ranitidine should be considered and used in a staggered manner. Erlotinib Tablets must be taken at least 2 hours before or 10 hours after the H2-receptor antagonist dosing.

Interaction with coumarin-derived anticoagulants, including warfarin, leading to increased International Normalized Ratio (INR) and bleeding events, which in some cases were fatal have been reported in patients receiving Erlotinib Tablets. Patients taking coumarin-derived anticoagulants should be monitored regularly for any changes in prothrombin time or INR.

The combination of Erlotinib Tablets and a statin may increase the potential for statin-induced myopathy, including rhabdomyolysis, which was observed rarely.

Smokers should be advised to stop smoking as cigarettes smoking, which is known to induce CYP1A1 and CYP1A2, has been shown to reduce erlotinib exposure by 50-60% (see sections, 2.2.2 Special Dosage Instructions, 3.2.5 Pharmacokinetics in Special Population).

In a phase Ib study, there were no significant effects of gemcitabine on the pharmacokinetics of erlotinib nor were there significant effects of erlotinib on the pharmacokinetics of gemcitabine.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS
3.1 Pharmacodynamic Properties
3.1.1 Mechanism of Action
Erlotinib potently inhibits the intracellular phosphorylation of HER1/EGFR receptor.
HER1/EGFR receptor is expressed on the cell surface of normal cells and cancer cells. In non-clinical models, inhibition of EGFR phosphotyrosine results in cell stasis and/or death.

3.1.2 Clinical / Efficacy Studies
Non-small cell lung cancer *(Erlotinib Tablets administered as monotherapy):*
First-line therapy for patients with EGFR activating mutations:
The efficacy of Erlotinib Tablets in first-line treatment of patients with EGFR activating mutations in NSCLC was demonstrated in a phase III, randomized, open-label trial (ML20650, EURTAC). This study was conducted in Caucasian patients with metastatic or locally advanced NSCLC (stage IIIB and IV) who have not received previous chemotherapy or any systemic antitumour therapy for their advanced disease and who present mutations in the tyrosine kinase domain of the EGFR (exon 19 deletion or exon 21 mutation). Patients were randomized 1:1 to receive Erlotinib Tablets 150 mg or platinum based doublet chemotherapy.

The primary endpoint of investigator assessed progression free survival (PFS), was determined at a pre-planned interim analysis (n=153, hazard ratio (HR)=0.42, 95% CI, 0.27 to 0.64; p<0.0001 for the Erlotinib Tablets group (n=77) relative to the chemotherapy group (n=0.76)). A 58% reduction in the risk of disease progression or death was observed. In the Erlotinib Tablets versus chemotherapy arms respectively, median PFS was 9.4 and 5.2 months and objective response rate (ORR) was 54.5% and 10.5%. PFS results were confirmed by an independent review of the scans, median PFS was 10.4 months in the Erlotinib Tablets group compared with 5.4 months in the chemotherapy group (HR=0.47, 95%CI, 0.27 to 0.78; p=0.003). The overall survival data were immature at the time of interim analysis (HR= 0.80, 95% CI, 0.47 to 1.37, p=0.4170).

At an updated analysis with 62% of OS maturity, OS HR was 0.93 (95 % CI, 0.64 to 1.36, p=0.7149). A high crossover was observed with 82% of the patients in the chemotherapy arm receiving subsequent therapy with an EGFR tyrosine kinase inhibitor and all but 2 of those patients had subsequent Erlotinib Tablets. In the updated analysis, PFS results remained consistent with the interim analysis results. Median PFS assessed by the investigators was 10.4 and 5.1 months in the Erlotinib Tablets and chemotherapy arms respectively (HR = 0.34, 95 % CI, 0.23 to 0.49, p<0.0001).

Additional published data
In a prospective analysis of patients with advanced NSCLC having tumours with activating mutations in the EGFR TK domain, the median PFS for the 113 patients treated with first-line Erlotinib Tablets was 14 months (95% CI, 9.7 to 18.3 months) and the median overall survival was 28.0 months (95% CI, 22.7 to 33 months).

A pooled analysis of published data from NSCLC patients showed that patients having tumours with EGFR activating mutations and receiving Erlotinib Tablets as predominantly firstline therapy (n=70, 12.5 months, 95%CI [10.6-16.0]) had a longer median PFS compared to those receiving chemotherapy (n=359, 6.0 months, 95%CI [5.4-6.7]).

First-line maintenance therapy:
The efficacy and safety of Erlotinib Tablets as first-line maintenance therapy of NSCLC was demonstrated in a randomized, double-blind, placebo-controlled trial BO18192 (SATURN). This study was conducted in 889 patients with locally advanced or metastatic NSCLC who did not progress during 4 cycles of platinum-based doublet chemotherapy. Patients were randomized 1:1 to receive Erlotinib Tablets 150 mg or placebo orally once daily. The primary end-point of the study was PFS in all patients and in patients with an EGFR IHC positive tumour. Baseline demographic and disease characteristics were well balanced between the two treatment arms.

In this study BO18192 (SATURN), the overall population showed a benefit for the primary PFS end-point (HR= 0.71 p< 0.0001) and the secondary OS end-point (HR= 0.81 p=0.0088). However the largest benefit was observed in a predefine exploratory analysis in patients with EGFR activating mutations (n= 49) demonstrating a substantial PFS benefit (HR=0.10, 95% CI, 0.04 to 0.25; p<0.0001) and an overall survival HR of 0.83 (95% CI, 0.34 to 2.02). 67% of placebo patients in the EGFR mutation positive subgroup received second or further line treatment with EGFR-TKIs. In patients with EGFR wild type tumors (n=388), the PFS HR was 0.78 (95% CI, 0.63 to 0.96; p=0.0185) and the overall survival HR was 0.77 (95% CI, 0.61 to 0.97; p=0.0243).

The BO25460 (IUNO) study was conducted in 643 patients with advanced NSCLC whose tumors did not harbor an EGFR-activating mutation (exon 19 deletion or exon 21 L858R mutation) and who had not experienced disease progression after four cycles of platinum-based chemotherapy.

The objective of the study was to compare the overall survival of first line maintenance therapy with erlotinib versus erlotinib administered at the time of disease progression.

The study did not meet its primary endpoint. OS of Erlotinib Tablets in first line maintenance was not superior to Erlotinib Tablets as second line treatment in patients whose tumor did not harbor an EGFR-activating mutation (HR= 1.02, 95% CI, 0.85 to 1.22, p=0.82). The secondary endpoint of PFS showed no difference between Erlotinib Tablets and placebo in maintenance treatment (HR=0.94, 95 % CI, 0.80 to 1.11; p=0.48).

Based on the data from the BO25460 (IUNO) study, Erlotinib Tablets use is not recommended for first-line maintenance treatment in patients without an EGFR activating mutation.

Second/Third-line therapy:
The efficacy and safety of Erlotinib Tablets in second/third-line therapy of NSCLC was demonstrated in a randomized, double-blind, placebo-controlled trial (BR.21). This study was conducted in 17 countries, in 731 patients with locally advanced or metastatic NSCLC after failure of at least one chemotherapy regimen. Patients were randomized 2:1 to receive Erlotinib Tablets 150 mg or placebo orally once daily. Study end points included overall survival, time to deterioration of lung cancer-related symptoms (cough, dyspnea and pain), response rate, duration of response, PFS, and safety. The primary end-point was survival.

Due to the 2:1 randomization, 488 patients were randomized to Erlotinib Tablets and 243 patients to placebo. Patients were not selected for HER1/EGFR status, gender, race, smoking history and histological classification.

Demographic characteristics were well balanced between the two treatment arms. About two-thirds of the patients were male and approximately one-third had a baseline ECOG performance status (PS) of 2 and 9% had a baseline ECOG of 3. Ninety-three percent and 92% of all patients in the Erlotinib Tablets and placebo groups, respectively, had received a prior platinum-containing regimen and 36% and 37% of all patients, respectively, had received a prior taxane therapy. Fifty percent of the patients had received only one prior regimen of chemotherapy.

Survival was evaluated in the intent-to-treat population. The median overall survival improved by 42.5% and was 6.7 months in the Erlotinib Tablets group (95% CI, 5.5 – 7.8 months) compared with 4.7 months in the placebo group (95% CI, 4.1 to 6.3 months). The primary survival analysis was adjusted for the stratification factors as reported at the time of randomization (ECOG PS, best response to prior therapy, number of prior regimens, and exposure to prior platinum) and HER1/EGFR status. In this primary analysis, the adjusted hazard ratio for death in the Erlotinib Tablets group relative to the placebo group was 0.73 (95% CI, 0.60 to 0.87) (p = 0.001). The percentage of patients alive at 12 months was 31.2% and 21.5%, respectively.

The survival benefit with Erlotinib Tablets treatment was seen across most subsets. A series of subsets of patients formed by the values of the stratification factors at randomization and at baseline, HER1/EGFR status, prior exposure to taxanes, smoking history, gender, age, histology, prior weight loss, time between initial diagnosis and randomization, and geographic location were examined in exploratory univariate analyses to assess the robustness of the overall survival result. Nearly all of the hazard ratios in the Erlotinib Tablets group relative to the placebo group were less than 1.0, suggesting that the survival benefit from Erlotinib Tablets was robust across subsets. Of note, the survival benefit of Erlotinib Tablets was comparable in patients with a baseline ECOG PS of 2-3 (HR = 0.77) or a PS of 0-1 (HR = 0.73) and patients who had received one chemotherapy regimen (HR= 0.76) or two or more regimens (HR=0.76).

A survival benefit of Erlotinib Tablets was also observed in patients who did not achieve an objective tumor response (by RECIST). This was evidenced by a hazard ratio for death of 0.83 among patients whose best response was stable disease and 0.85 among patients whose best response was progressive disease.

Table 3 summarizes the results for the study, including survival, time to deterioration of lung cancer-related symptoms, and progression-free survival (PFS).

Table 3 Study BR.21 Efficacy Results

	Erlotinib Tablets (N=488)	**Placebo (N=243)**	**p-value**
Median survival 95% CI	6.7 months (5.5 to 7.8)	4.7 months (4.1 to 6.3)	
Difference between survival curves			0.001
Hazard ratio*, mortality (erlotinib: placebo) 95% CI (risk ratio)	0.73 0.60 to 0.87		
Median time to deterioration in cough*** 95% CI	28.1 weeks (16.1 to 40.0)	15.7 weeks (9.3 to 24.3)	0.041
Median time to Deterioration in Dyspnea*** 95% CI	20.4 weeks (16.3 to 28.3)	12.1 weeks (9.3 to 20.9)	0.031**
Median time to deterioration in pain*** 95% CI	12.1 weeks (10.1 to 14.1)	8.1 weeks (7.7 to 12.3)	0.040**
Median progression-free survival 95% CI	9.7 weeks (8.4 to 12.4)	8.0 weeks (7.9 to 8.0)	<0.001

*Adjusted for stratification factors and HER1/EGFR status; a value less than 1.00 favors Erlotinib Tablets (primary analysis).
**p-value adjusted for multiple testing.
***From the EORTC QLQ-C30 and QLQ-LC13 Quality of Life Questionnaire.
Symptom deterioration was measured using the EORTC QLQ-C30 and QLQ-LC13 quality of life questionnaires. Baseline scores of cough, dyspnea and pain were similar in the two treatment groups. Erlotinib Tablets resulted in symptom benefits by significantly prolonging time to deterioration in cough (HR =0.75), dyspnea (HR=0.72) and pain (HR=0.77), versus placebo. These symptom benefits were not due to an increased use of palliative radiotherapy or concomitant medication in the Erlotinib Tablets group.

The median PFS was 9.7 weeks in the Erlotinib Tablets group (95% CI, 8.4 – 12.4 weeks) compared with 8.0 weeks in the placebo group (95% CI, 7.9 to 8.1 weeks). The HR for progression, adjusted for stratification factors and HER1/EGFR status, was 0.61 (95% CI, 0.51 to 0.73) (p < 0.001). The percent of PFS at 6 months was 24.5% and 9.3%, respectively, for the Erlotinib Tablets and placebo arms.

The objective response rate by RECIST in the Erlotinib Tablets group was 8.9% (95% CI, 6.4 to 12.0%). The median duration of response was 34.3 weeks, ranging from 9.7 to 57.6+ weeks. Two responses (0.9%, 95% CI, 0.1 to 3.4) were reported in the placebo group.

The proportion of patients who experienced complete response, partial response or stable disease was 44.0% and 27.5%, respectively, for the Erlotinib Tablets and placebo groups (p=0.004).